

Q1. Autonomous Vehicle Navigation Problem Statement: Design an RL agent that can safely navigate an autonomous vehicle through a simulated urban environment. The agent must learn to follow traffic rules, avoid collisions, and reach its destination in the shortest time possible.

Q2. Smart Grid Energy Management Problem Statement: Develop an RL agent that optimizes energy consumption in a smart grid, balancing the supply and demand by dynamically pricing energy based on usage patterns, weather conditions, and renewable energy availability.

Q3. Personalized Education Problem Statement: Create an RL-based system that personalizes learning experiences for students by adapting the difficulty level and topics of educational content based on the student's performance and engagement.

Q4. Healthcare Treatment Optimization Problem Statement: Design an RL agent that suggests personalized treatment plans for patients with chronic conditions, such as diabetes, by continuously learning from patient data, treatment outcomes, and evolving medical research.

Q5. Frozen Lake Navigation Problem Statement: The agent must learn to navigate across a frozen lake from a starting point to a goal, avoiding falling into holes. This problem is a classic example of a discrete state and action space, ideal for introducing Q-learning. Use the Q-learning algorithm to find the best action in each state. The environment provides a reward of 1 if the agent reaches the goal and 0 otherwise.

Q6. Understanding Rewards and Actions:

An RL agent is navigating a grid world, a simple environment used for demonstration purposes. The grid is 4x4, and the agent starts in the top left corner (0,0) and must reach the bottom right corner (3,3) to maximize its cumulative reward. The actions it can take at any state are to move up, down, left, or right. Moving into a wall keeps the agent in the same place.

The agent receives a reward of -1 for each action taken and +10 for reaching the goal. If the agent takes a path right, right, down, down, down, right, calculate the total cumulative reward. Discuss how changing the reward for reaching the goal to +20 would affect the agent's learning.

Q7. Discount Factor in Future Rewards:

Consider the same grid world. Now, introduce a discount factor (γ) that determines the importance of future rewards. A common value for γ is 0.9. Assume the agent takes the following actions: right, right, down, down, right, and then reaches the goal. Calculate the total cumulative reward for the agent using a discount factor of 0.9. Explain the effect of changing the discount factor to 0.99 on the agent's strategy.

Q8. Exploration vs. Exploitation Trade-off:

An RL agent is playing a slot machine with two levers. Lever A returns a reward of 1 with a probability of 0.5, and Lever B returns a reward of 2 with a probability of 0.2. The agent has a policy to explore (choose randomly) 20% of the time and exploit (choose the best lever based on past experience) 80% of the time. If the agent pulls a lever 100 times, estimate the expected

total reward. Discuss how the agent's total expected reward might change if it explores 50% of the time and exploits 50% of the time.

Q9. Discount Factor in Future Rewards:

Consider the same grid world. Now, introduce a discount factor (γ) that determines the importance of future rewards. A common value for γ is 0.9. Assume the agent takes the following actions: right, right, down, down, right, and then reaches the goal. 1. Calculate the total cumulative reward for the agent using a discount factor of 0.9. 2. Explain the effect of changing the discount factor to 0.99 on the agent's strategy.

Q10. Exploration vs. Exploitation Trade-off:

An RL agent is playing a slot machine with two levers. Lever A returns a reward of 1 with a probability of 0.5, and Lever B returns a reward of 2 with a probability of 0.2. The agent has a policy to explore (choose randomly) 20% of the time and exploit (choose the best lever based on past experience) 80% of the time. If the agent pulls a lever 100 times, estimate the expected total reward. Discuss how the agent's total expected reward might change if it explores 50% of the time and exploits 50% of the time.

Q11. Q-Learning Basic Calculation:

An agent is learning to navigate the grid world using Q-learning, a popular RL algorithm. The Q-value $Q(s,a)$ represents the quality of taking action a in state s . Assume the agent is in state (2,2), and the Q-values for moving left and right are 0.5 and -0.2, respectively. After taking the action to move left, the agent receives a reward of -1 and ends up in state (2,1) with the highest Q-value for the next actions being 0.7. The learning rate (α) is 0.1, and the discount factor (γ) is 0.9. Calculate the updated Q-value for the action taken (moving left from state (2,2)). Explain the significance of the learning rate and discount factor in the Q-learning update formula.

Q12. Policy Improvement:

An RL agent uses a policy π to choose actions in a simplified environment with three states (A, B, C) and two actions (X, Y). The transition rewards are as follows: A-X leads to B with a reward of +2, A-Y leads to C with a reward of +1, B-X leads to C with a reward of +2, B-Y leads to A with a reward of -1, C-X leads to A with a reward of +0, and C-Y leads to B with a reward of +3. Assuming the agent starts with a policy to always choose action X, calculate the total reward for two transitions starting from each state. Discuss how the agent could improve its policy based on the rewards received from the transitions.

Q13. Autonomous Driving:

An autonomous vehicle is trained using RL to navigate through a city environment. The vehicle must learn to reach a destination as quickly as possible while avoiding obstacles and obeying traffic laws. The state space includes the vehicle's speed, direction, location, and the proximity of obstacles. Actions include accelerating, decelerating, and turning. The reward function gives positive points for moving closer to the destination, negative points for collisions, and penalties for traffic law violations. Design a reward function that encourages the vehicle to reach its destination efficiently while ensuring safety and law compliance. Discuss how you would

implement an exploration strategy that allows the vehicle to learn about its environment without compromising safety.

Q14. Healthcare Treatment Optimization:

An RL agent is used to recommend personalized treatment plans for patients with chronic diseases, considering the patient's health state, medical history, and treatment response. The state space includes various health indicators, treatment history, and patient-reported outcomes. Actions include adjusting medication dosages, changing medications, or recommending lifestyle changes. The reward function aims to improve patient health outcomes while minimizing side effects and costs. Propose a reward function that accurately reflects the goal of optimizing patient health outcomes. Discuss the ethical considerations and the importance of safety in exploring new treatment plans with RL.

Q15. Policy Evaluation in Grid World Scenario:

Consider a simplified 3x3 grid world where an agent can move in four directions: up, down, left, and right. The agent starts in the top left corner (0,0) and aims to reach the bottom right corner (2,2) with the least number of steps. All actions have a deterministic outcome. The agent receives a reward of -1 for each step and +10 for reaching the goal. Assume we have a policy π that chooses each action with equal probability. Calculate the expected return from each state under policy π using a discount factor (γ) of 0.9. How would the expected return change if the policy is updated to always choose the action that moves directly toward the goal?

Q16. Bellman Equation for Value Estimation:

In the same 3x3 grid world, consider a state (1,1) with possible actions leading to states (1,0), (1,2), (0,1), and (2,1). The immediate rewards for moving to these states are -1, and the discount factor γ is 0.9. Assume the current estimated values of these states are 2.0, 1.0, 0.5, and 3.0, respectively. Use the Bellman equation to calculate the updated value of state (1,1) if the agent follows a greedy policy that always chooses the action leading to the state with the highest value. Discuss the impact of changing the discount factor to 0.99 on the value of state (1,1).

Q17. Q-Learning Update Scenario:

An agent is learning to play a simple game using Q-learning. The agent is in state S and chooses action A, leading to state S'. The reward received is $R = 2$. The learning rate α is 0.1, and the discount factor γ is 0.95. The highest Q-value for the next state S' is 5. If the current Q-value $Q(S,A)$ is 3, calculate the updated Q-value after taking action A and receiving the reward. Explain how the Q-value would change if the reward for taking action A was increased to $R = 5$